

- **WHAT IS DATA  
SCIENCE?**

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# AGENDA

introduction-what is data science?

The data deluge-why data science is essential today

When did data science come to the world?

Where data science is used

Why data science is used

How data science is used

Pros and cons

Examples and practical points

The future of data science-trends and opportunities

# WHAT IS DATA SCIENCE?

Data science is the multidisciplinary practice of extracting insights from data using statistics, algorithms and domain knowledge. It blends programming, math and communication to turn raw information into decisions and products.



# THE DATA DELUGE-WHY DATA SCIENCE IS ESSENTIAL TODAY

Everyday systems generates massive volumes of data-sensors, transactions, logs and medias. data science makes this sea of information actionable: detecting patterns, forecasting outcomes and automating decisions.

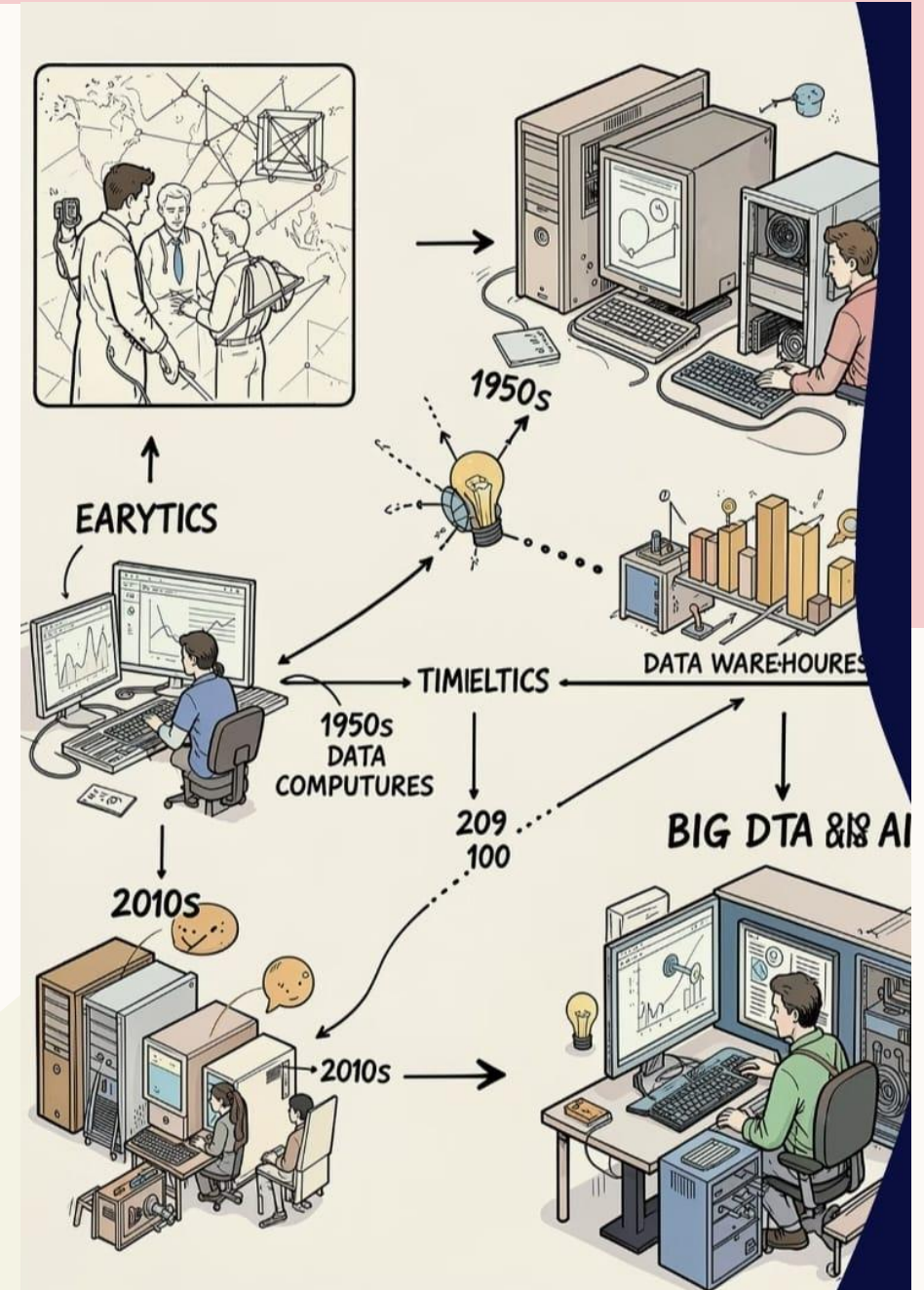
Three majors are :

- \*scale
- \*speed
- \*variety



# WHEN DID DATA SCIENCE COME TO THE WORLD?

Data science evolved gradually: statistical methods came first, computing enabled scale, warehouses enabled enterprise analytical, and the big data era plus machine learning created the modern field

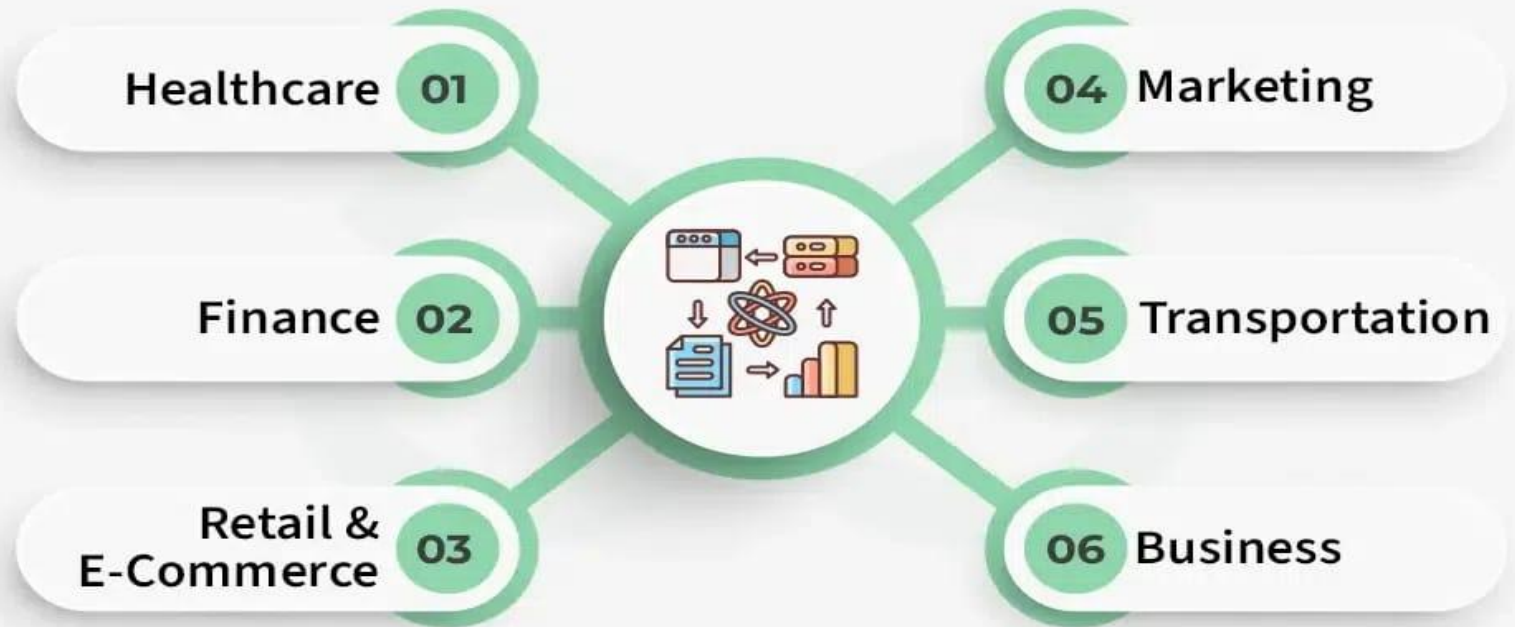


# WHERE DATA SCIENCE IS USED

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## The Role of Data Science in Everyday Life



# WHY DATA SCIENCE IS USED

Make better decisions using evidence and prediction

Automate repetitive decision at scale discover hidden patterns and new opportunities personalize products and customer experiences reduce risk through forecasting and anomaly detection

Efficiency

Optimize processes, cut costs

Growth

Identify new markets and upshell paths

Innovation

Product features powered by ML

# HOW DATA SCIENCE IS DONE

1. TYPICAL WORKFLOW: collect data from sources, clean and preprocess, explore with visualization, built and validate models, then deploy and monitor in production for feedback and improvement

Collect: gather raw data from sources

Clean: preprocesses and fix data issue

Explore: visualize patterns and insights

Models: built and validate predictive models

Deploy: release and monitor in production



# PROS AND CONS OF DATA SCIENCE

- PROS
  - Data-driven decisions
  - Automation at scale
  - Competitive advantages
- CONS
  - Data bias&fairness risks
  - Privacy and compliance challenges
  - High infrastructure and talent costs

# EXAMPLES & PRACTICAL POINTS

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Recommendation  
systems

Churn prediction

Computer vision in  
healthcare

Additional  
points: reproducibility,  
versioning data\models



# THE FUTURE OF DATA SCIENCE: TRENDS & OPPORTUNITIES

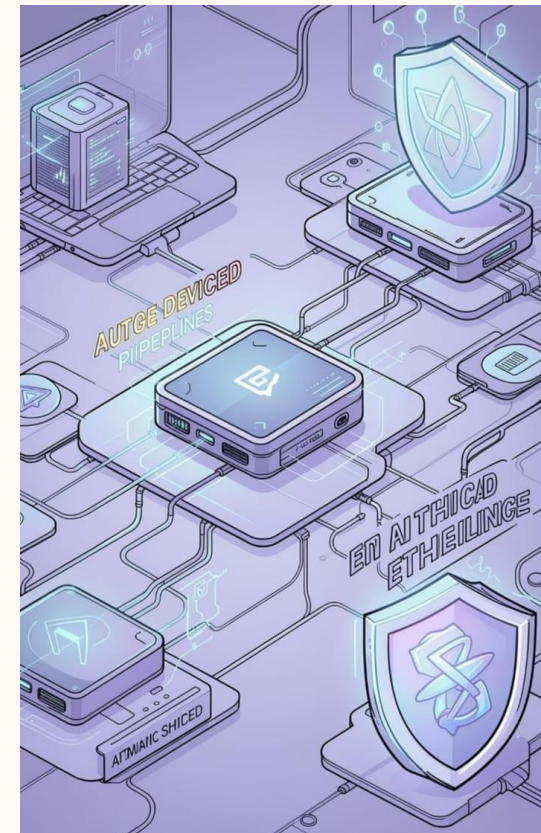
autoML and low code tools accelerating model development

Edge & real-time analytics for low-latency application

Responsible AI with stronger governance and explainability

Greater integration of domain knowledge and casual inference

Growing demand for cross-disciplinary practitioners (MLOps, data engineering)



**THANK  
YOU**